Assignment No. 01

Que 1

a) jshell > int a = 'A'; a+4

⇒ Output : 69

Explanation: ASCII Value of 'A' is = 65

65 + 4

69

b) Assume a = 5

jshell > -a - ++a + --a

⇒ output : -5

Explanation :

-a → -5

++a → 6 (a becomes 6)

- -a → -(-6) → +6

-5 - (6) + 6

= -5

c) jshell > String b = "A\tb c\sd"

jshell > b.replace(" ","")

⇒ Output : ""A\tbcd"

Explanation: \t = tab , \s = space.

replace(" ","") removes only spaces.

not tab

"A\tbcd".

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d) jshell > Byte.SIZE + short.SIZE / short.Bytes

→ Error

~~Ter~~ Explanation: Java is case-sensitive
It cannot find symbol : short.Bytes.

The correct constant name is
Short.BYTES

Que 2

a)

→ Print X pattern for odd size, else show message "Invalid Number Enter Odd No"

Code

```
import Java.util.* ;

class pattern {
    void printpattern (int size) {
        if (size % 2 == 0) {
            System.out.println (" Invalid Number,
                please Enter odd Number") ;
            return ;
        }
        for (int i=0 ; i< size ; i++) {
            for (int j=0 ; j<=size ; j++) {
                if (j==i || j == size-i-1) {
                    System.out.print (" X") ;
                }
```


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```
            else {
                System.out.print(" ");
            }
        }
        System.out.println();
    }
}
public static void main(String[] args){
    ~~void~~ printPattern(5);
}
```

output

```
X       X
  X   X
    X
  X   X
X       X
```


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Que 2 b)

Evaluate the implications of explicitly casting large integers to the short type in Java, using (short) 5022026 as an example

→

short = 16-bit

Signed range -32768 to 32767

Total values = 65536 (2^{16})

When a large int is cast to short, Java keeps only the lower 16 bits and discard higher bits (overflow occurs)

5022026 mod 65536 = 41290

41290 - 65536 = -24246

Ans : (short) 5022026 = -24246

Que 2 c)

Can the isBlank() method of the String class always be used as replacement for isEmpty()? Justify your answers with suitable example.

→ No

isEmpty() → checks length = 0 only

isBlank() → checks empty or only whitespace

e.g. String s = " ";

System.out.println(s.isEmpty()); // false

System.out.println(s.isBlank()); // true.

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